Two lanes, one corridor: independent packaging as a test of admissibility structure in stress-weighted pedestrian routing

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April 2026

Abstract

We constructed two independent figure-generation pipelines—**pax** and **synapsin**—for a stress-weighted pedestrian routing system on a fixed midtown Manhattan corridor (Streets 42–59, Avenues 3–9) under the cost model $\text{cost}(e) = d_e + \alpha\sigma_e$ with $\alpha = 0.088$. Both pipelines read the same frozen laboratory outputs; neither was edited to match the other. The pax pipeline carries the full analysis chain: a frozen admissibility registry (six entries), regime adjacency, divergence evaluation, a paired cost-semantic trace experiment (OD 553 → 486), and mechanical verification. The synapsin pipeline renders the admissibility topology tables through a separate path. Both arrive at the same registry structure and adjacency outcomes. Where they organize evidence differently—synapsin attaches admissibility figures to input-closure claims while pax routes those claims to a hydration manifest—the divergence reflects editorial choice, not missing data. A paired-trace cost experiment yields a four-edge shared suffix with zero per-hop cost delta and a total cost difference of ≈0.121 between unregularized and regularized legs, confirmed by mechanical verification. The paper presents these results in three groups: convergence, designed asymmetry, and depth available only in the pax pipeline.

1 Introduction

A pedestrian routing system that accounts for perceptual stress alongside geographic distance must justify its cost model, its spatial inputs, and the evaluation chain that connects them. The system described here operates on a graph of 554 intersections in a midtown Manhattan corridor, with edge costs defined as $\text{cost}(e) = d_e + \alpha\sigma_e$, where d_e is the Euclidean edge distance, σ_e is an interpolated stress proxy derived from traffic camera imagery, and $\alpha = 0.088$. The corridor envelope is fixed (Streets 42–59, Avenues 3–9), the graph identity is frozen, and the cost binding is immutable for the work reported here.

Rather than present the analysis through a single pipeline, we constructed two independent figure-generation paths—**pax** and **synapsin**—that take the same frozen laboratory outputs and render them separately. The question is not whether the system produces a particular routing outcome, but whether its evidentiary chain is stable when read through two paths that were not written to agree.

Where both paths show the same structure, convergence is meaningful. Where they organize evidence differently, the divergence reveals how editorial choices about attaching evidence to argument can vary without contradiction. Where only one path carries a particular figure, the asymmetry is informative depth, not a gap.

2 System and cost model

The routing system derives stress scores from approximately 14,800 images collected across 82 New York City Department of Transportation traffic cameras. Multi-scale feature extraction

(object detection, scene embeddings, segmentation) yields camera-level stress proxies grounded in pedestrian density, spacing, and infrastructure context. Interpolation by inverse distance weighting maps camera-zone scores to intersection-level values across the corridor graph.

Temporal structure in this dataset is strong: a 12-basis raised-cosine kernel fits the diurnal stress cycle with $R^2 = 0.978$, rush-hour stress exceeds off-peak by approximately 5% (Hedges' $g = 0.17$, $p < 10^{-27}$), and weekend stress falls roughly 6% below weekday levels ($g = -0.20$, $p < 10^{-41}$). Spatial discrimination between avenues, by contrast, is not statistically distinguishable at the camera level (e.g. Park vs. Madison $g \approx -0.01$, confidence interval spanning zero). The routing layer uses calibrated parameters—including domain-knowledge boosts and discounts for specific corridors—rather than learned spatial preferences; it is a demonstration of the cost framework, not a claim of feature discovery.

Edge costs follow $\text{cost}(e) = d_e + 0.088\,\sigma_e$ throughout. The cost binding, corridor envelope, and graph identity are treated as immutable inputs for all work below.

3 Two-pipeline design

Both pipelines draw from the same set of frozen laboratory files: CSV tables for registry freeze summaries, regime adjacency matrices, divergence evaluation rows, suffix edge ladders, per-hop cost tables, and aggregate evaluation quantities. Each pipeline independently reads these files and produces its own figures and metadata.

The **pax** pipeline carries the full figure set: admissibility structure (Figure 1), cost-semantic trace (Figure 3), verification closure (Figure 4), evaluation pipeline (Figure 5), stage-zero closed inputs (Figure 2), and a semantic-gate artifact (Figure 6).

The **synapsin** pipeline renders only the registry freeze summary and regime adjacency panels from the same inputs, using a separate rendering script and a narrower metadata format.

Comparison between the two pipelines uses the shared evidence identifiers (claim and panel labels) as join keys. Where both pipelines produce a figure for the same identifier, they can be placed side by side. Where one pipeline attaches a figure to a claim that the other handles differently, the relationship is functional rather than referential.

Neither pipeline was edited to match the other. What follows is what the figures show when read together.

4 Shared invariants

Six frozen admissibility entries (V1–V6) appear in both renderings (Figure 1), each pointing to a boundary registry artifact from the 2026-04-02 laboratory session. The transition surface reference and all interior regime hashes are identical across renderings.

Regime adjacency (Figure 1c) tabulates pairwise adjacency among the six interior regimes; both pipelines reproduce the same outcomes.

Divergence evaluation (Figure 1a, panel F1C) reports the ratio of divergence step index to path length in edges for each entry. All six rows classify as *early* under the one-third / two-thirds partition: V1 and V5 show a ratio of 0.0 (divergence at step 0), while V2–V4 and V6 range from approximately 0.03 to 0.12.

These tables record which regimes exist, how they neighbor each other, and how early divergence occurs—the structural precondition for cost evaluation, not its result. That two independent pipelines render the same precondition from the same frozen inputs, without coordination, is the first point of convergence.

Figure placeholder

Removed in the 2026-04-16 review build because the rendered visualization was broken.

Placeholder for pax rendering: registry freeze summary (F1A), regime adjacency (F1B), and divergence evaluation (F1C).

(a) pax rendering: registry freeze summary (F1A), regime adjacency (F1B), and divergence evaluation (F1C).

Figure placeholder

Removed in the 2026-04-16 review build because the rendered visualization was broken.

Placeholder for synapsin F1-A: registry freeze summary with doctrine identity and cost formulation fields.

(b) synapsin F1-A: registry freeze summary with doctrine identity and cost formulation fields.

Figure placeholder

Removed in the 2026-04-16 review build because the rendered visualization was broken.

Placeholder for synapsin F1-B: regime adjacency.

(c) synapsin F1-B: regime adjacency.

Figure 1: Two independent renderings of the admissibility topology converge on the same registry entries (V1–V6), regime adjacency outcomes, and divergence ratios. Top: pax rendering (F1A registry, F1B adjacency, F1C divergence). Bottom: synapsin rendering of F1-A and F1-B, which additionally displays the cost formulation and $\alpha = 0.088$ inline. Neither rendering was coordinated with the other.

5 Structural divergence

Both pipelines attach admissibility figures to claims about bounded extraction and pipeline structure. But they disagree on how to handle claims about input closure and promotion integrity. Synapsin attaches the same registry and adjacency tables to those claims; pax instead points to a stage-zero hydration manifest (Figure 2) that enumerates the 24 closed inputs resolved before any figure was drawn.

This is not a gap. It is two pipelines making different editorial decisions about what "the inputs are closed" looks like as a figure. Synapsin argues that the admissibility tables themselves already demonstrate closure. Pax argues that closure is better shown by the manifest of what was resolved before any figure was drawn. Both are defensible, and the fact that both exist without contradiction makes the divergence informative: it shows that evidence-to-claim attachment is a design choice, not a mechanical given.

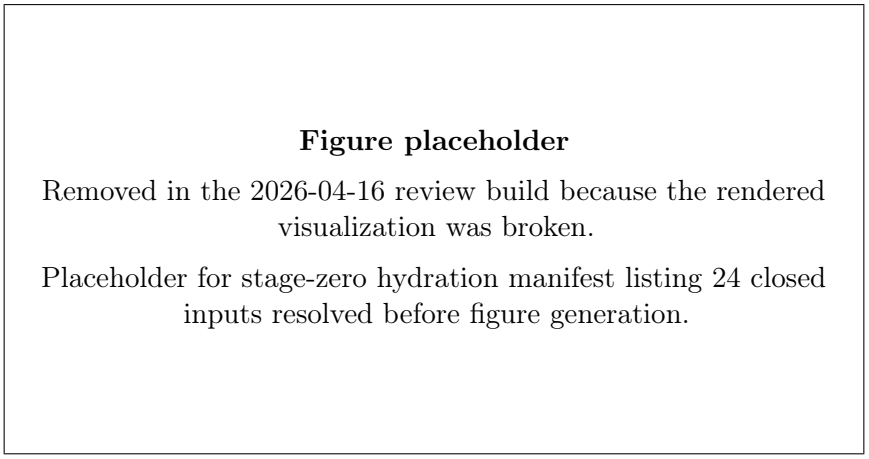


Figure 2: Input closure evidence as shown by pax: a stage-zero hydration manifest listing 24 closed inputs resolved before figure generation. Synapsin handles the same claim by re-using the admissibility tables (Figure 1). The two pipelines attach the same claim to different evidence, and both are defensible.

6 Cost-semantic trace

The cost experiment pairs two routes on OD $553 \to 486$: leg A without regularization and leg B with it enabled. Intake validation reports a pass. The two routes share a four-edge terminal suffix: $(558 \to 165)$, $(165 \to 170)$, $(170 \to 513)$, $(513 \to 486)$.

On every hop of this shared suffix, per-hop edge costs match exactly across legs (Figure 3, top and middle panels): the maximum absolute per-hop delta is 0.0. The entire cost difference between the two routes arises in the prefix—the portion of each path before the routes converge onto their shared terminal approach.

Terminal cumulative path costs are $0.5145\ldots$ for leg A and $0.3934\ldots$ for leg B, yielding a total cost delta of approximately 0.121 and a relative delta of roughly 30.8% over leg B. Because suffix costs are identical hop by hop, the prefix cost delta at the suffix entry point equals the total delta: switching regularization changes how the route reaches the shared suffix, but not the terminal convergence.

This is one OD pair with one binary knob. It does not generalize to other origins, destinations, or parameter ranges without further experiments. What it demonstrates is that the evaluation procedure can isolate *where* in a paired trace the cost difference arises and that the resulting quantities are mechanically verifiable.

Synapsin does not carry cost-trace figures; it was scoped to admissibility topology. The cost evidence is pax-only, and it functions as depth within the synthesis, not as an absent counterpart.

7 Verification and pipeline closure

The verification script (Figure 4) reads the two trace reports, aligns the terminal suffix, recomputes per-hop deltas and cumulative costs, and reports pass or fail. It operates entirely on serialized tables and does not re-run the router. The result is `pass` with no discrepancy, confirming that the quantities in Figure 3 are reproducible from the archived artifacts alone.

Figure 5 shows the dataflow: two trace-leg artifacts feed a cost evaluation step, which feeds the verification script. The aggregate cost quantities are not self-reported summaries; they are outputs of a reproducible alignment procedure with an independent witness.

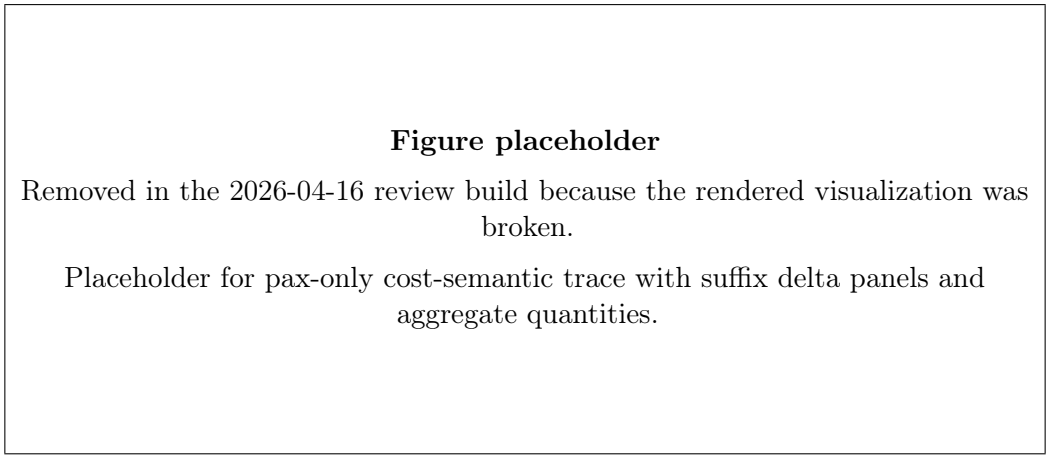


Figure 3: All cost difference between the unregularized and regularized routes arises before the paths converge (pax only). **Top**: per-hop cost delta on the four-edge shared suffix is exactly zero. **Middle**: tabulated confirmation. **Bottom**: aggregate quantities—interpret alongside the verification witness in Figure 4. Cumulative costs: 0.5145 (leg A) vs. 0.3934 (leg B); total delta ≈0.121; relative delta ≈30.8%.

8 Semantic gate and provenance discipline

A semantic gate (Figure 6) governs what kinds of claims topology-only analyses can support. Any procedure that compares node sequences, divergence indices, or regime adjacency *without* reading a cost-layer field (d_e, σ_e, or edge cost) is classified as provenance-only for cost-semantic claims, even when the underlying traces come from cost-bound runs.

Under this rule, four of six registry entries (V1–V4) pass whole-stack validity; V5 and V6 are provenance-only. The distinction prevents the admissibility tables in Section 4 from being read as standalone proof of cost outcomes. Those tables show what the registry contains; the cost-trace figures in Section 6 show what the costs do. The semantic gate keeps those two readings separate.

9 Discussion

The reading order reflects the composability of the two pipelines, not a presentational convenience.

Shared invariants (Section 4) establish that registry structure and regime adjacency are stable properties of the frozen inputs. Two pipelines, built without coordination, render the same tables from the same files. This convergence is the baseline against which the subsequent sections become interpretable.

Structural divergence (Section 5) shows that evidence-to-claim attachment is editorial. The same underlying data can support "the inputs are closed" through a hydration manifest (pax) or through the admissibility tables themselves (synapsin). Neither choice is wrong; the coexistence of both without contradiction is itself a finding about how evidence structures can vary within a single analytical program.

The pax-only sections (Sections 6–8) show what one pipeline can carry when it is not constrained to match a counterpart. The cost-trace experiment, verification, pipeline diagram, and semantic gate constitute the evidentiary core for cost claims. Their absence from synapsin reflects scope, not omission: synapsin stays focused on admissibility topology while pax carries cost evaluation and closure.

Limitations. The cost experiment covers one OD pair (553 → 486) with one binary knob. It does not generalize to other origins, destinations, or parameter ranges without further experiments.

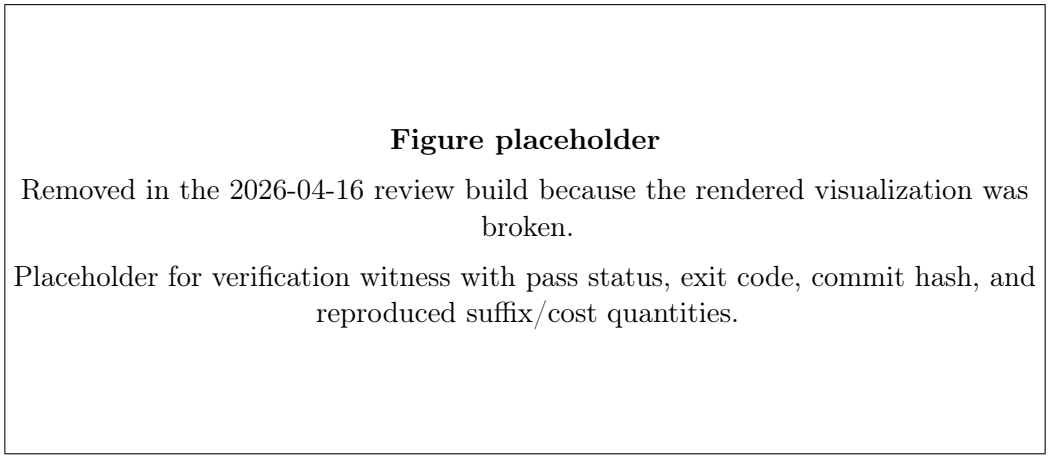


Figure 4: Independent witness for the cost-trace quantities in Figure 3. Left: verification metadata showing pass status, exit code 0, commit hash, and no discrepancy. Right: log tail reproducing the shared suffix, per-hop deltas (all 0.0), cumulative costs, and prefix delta. The verification script reads only serialized trace reports; it does not recompute routes.

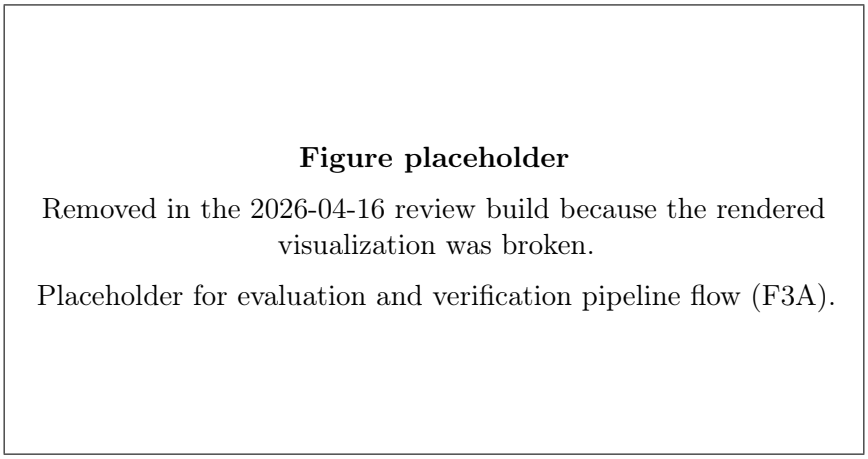


Figure 5: Evaluation and verification pipeline (F3A). The flow reads left to right: two trace-leg artifacts feed a cost-semantic evaluation step, which feeds the mechanical suffix-and-Q verification script. The validation report is cited in the manuscript path but not redrawn here.

Spatial stress discrimination between avenues is not statistically distinguishable in the camera data; routing preferences for specific corridors reflect calibrated domain knowledge, not learned weights. The interpolation pipeline explains approximately 28% of spatial variance in leave-one-camera-out cross-validation. Topology-only admissibility metrics are provenance-only for cost claims per the semantic-gate rule.

What this paper does not claim. It does not claim routing optimality, stress causality, or corridor expansion beyond the stated envelope. It does not merge topology-only metrics with cost quantities in the same assertion. It does not assert that the two pipelines produce byte-identical outputs; it asserts that they arrive at the same structural conclusions from the same frozen inputs through independent paths.

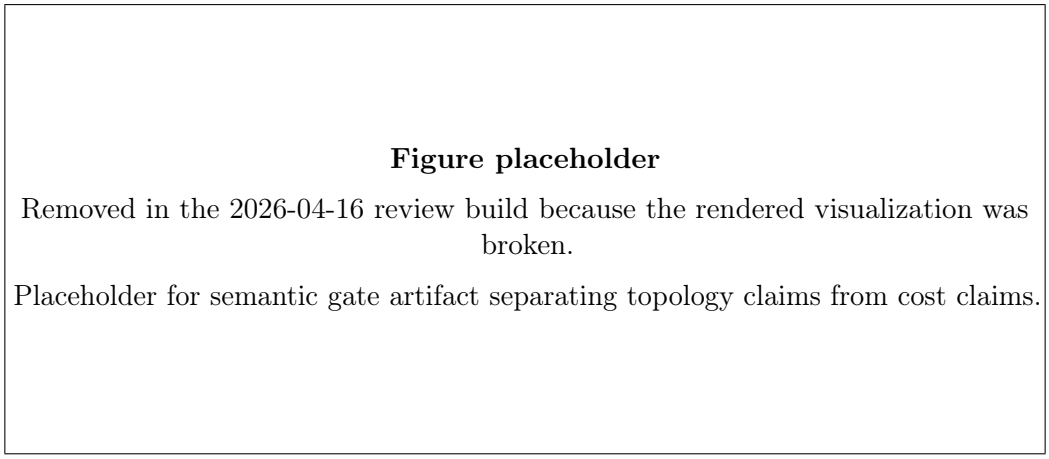


Figure 6: Semantic gate: the rule that separates topology claims from cost claims. Left: gating status showing V1–V4 as whole-stack valid and V5–V6 as provenance-only. Right: the gate condition—a procedure must read a bound cost-layer field (d_e, σ_e, or edge cost) to support cost-semantic claims. This figure documents provenance discipline; it does not itself carry cost evidence.

10 Conclusion

Two independently constructed pipelines, reading the same frozen laboratory outputs under a fixed cost model, converge on admissibility topology and diverge in how they attach that topology to downstream claims. The convergence is evidence that registry structure and regime adjacency are stable properties of the data. The divergence is evidence that evidence-to-claim attachment admits more than one defensible resolution. The pax-only cost trace, verification, and pipeline figures supply the evidentiary depth for cost-semantic readings without requiring the second pipeline to duplicate them.

Taken together, the synthesis shows that the evidentiary chain for this routing system is reproducible, that its internal organization tolerates independent editorial decisions, and that its cost evaluation is mechanically verifiable from archived artifacts.